ASNet: Adaptive Semantic Network Based on Transformer–CNN for Salient Object Detection in Optical Remote Sensing Images

Ruixiang Yan, Longquan Yan, Guohua Geng, Yufei Cao, Pengbo Zhou, and Yongle Meng

***Abstract*— Salient object detection in optical remote sensing images (RSI-SOD) has recently become a key area of research, driven by the unique challenges posed by specific imaging conditions. Traditional approaches, largely based on convolutional neural networks (CNNs), are limited in handling the diverse scenarios of remote sensing due to their static network construction and reliance on local feature extraction. To tackle these limitations, we present the adaptive semantic network (ASNet), a novel framework specifically designed for RSI-SOD. ASNet innovatively integrates Transformer and CNN technologies in a dual-branch encoder, which captures both global dependencies and local fine-grained image details. The network also features an adaptive semantic matching module (ASMM) for dynamically harmonizing filter responses to global and local contexts, an adaptive feature enhancement module (AFEM) that effectively enhances salient region features while restoring image resolution, and a multiscale fine-grained inference module (MFIM) that refines high-level semantic features by integrating detailed low-level information, leading to the generation of precise, high-quality saliency maps. These components work in concert to adaptively respond to the complex nature of remote sensing images (RSIs). Extensive experimental evaluations confirm that ASNet substantially outperforms existing models in the RSI-SOD task.**

***Index Terms*— Adaptive semantics, dynamic convolution, optical remote sensing images (RSIs), salient object detection (SOD), Transformer.**

I. Introduction

THE human visual attention mechanism is adept at discerning the most salient objects or regions within an image, a concept central to the study of salient object detection (SOD) [1]. This field aims to precisely locate and accurately segment the most visually striking objects across diverse scenes. SOD supports a variety of downstream tasks such as object recognition [2], segmentation [3], [4], tracking [5], [6], image retrieval [7], retargeting [8], and quality assessment [9], [10]. The last decade has seen considerable advancements in SOD in natural scene images (NSI-SOD), leading to various research directions, including video SOD [11], RGB-D SOD [12], [13], [14], and co-saliency detection [15]. Of particular interest is SOD in optical remote sensing images (RSI-SOD), an emerging field that has attracted considerable academic attention.

In line with the evolution of SOD, significant strides have been made in remote sensing technology, particularly in hyperspectral image classification and semantic segmentation of optical remote sensing images (RSIs). In this realm, the works of Zhang et al. [16], [17] are noteworthy. Their single-source domain expansion network (SDEnet) employs generative adversarial learning, introducing a novel approach for domain generalization in cross-scene hyperspectral image classification. They also developed the topological structure and semantic information transfer network (TSTnet), aligning topological relationships in hyperspectral images to boost cross-scene classification accuracy. Complementing these, Zhang et al. [18] explored morphological transformations and spatial-logical aggregation for tree species classification, leveraging morphological operators to extract finer spatial features. In addition, Zhou et al.'s [19] DSM-assisted unsupervised domain adaptive Network and Li et al.'s [20] semantic-preserved generative adversarial network (SPGAN-DA) mark significant advancements in semantic segmentation, addressing challenges in domain adaptation. These recent advancements in remote sensing technology offer valuable insights and serve as important references for the continued development of RSI-SOD research.

RSI-SOD has significant implications across various practical applications, profoundly impacting urban planning, environmental monitoring, agriculture, disaster management, defense, and resource exploration [21], [22], [23]. Its role in urban planning includes identifying key features such as buildings and roads in RSIs, while in environmental monitoring, it assists in tracking land use changes and ecosystem health. In agriculture, SOD facilitates precision farming by analyzing crop types and conditions. Its importance extends to disaster management, aiding in rapid response through landscape change detection, and to defense and surveillance for monitoring borders and detecting suspicious activities.

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Ruixiang Yan, Longquan Yan, and Guohua Geng are with the School of Information Science and Technology, Northwest University, Xi'an 710119, China (e-mail: yanrx5@163.com; 18829512640@163.com; ghgeng@nwu.edu.cn).

Yufei Cao is with the School of Information Engineering, Ningxia University, Yinchuan 750021, China (e-mail: cc_yff@163.com).

Pengbo Zhou is with the College of Information Science and Technology, Beijing Normal University, Beijing 100875, China (e-mail: zhoupengbo@bnu.edu.cn).

Yongle Meng is with the School of Chemistry and Chemical Engineering, Xi'an University of Science and Technology, Xi'an 710054, China (e-mail: myl15277116706@163.com).

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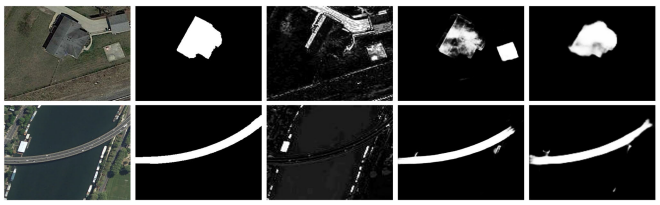


Fig. 1. Challenging optical remote sensing scenes with saliency maps inferred by three classic methods. The figure illustrates the visual limitations of traditional SOD methods (e.g., VOS [34]) and mature NSI-SOD methods (e.g., GateNet [31]) on RSIs. In contrast, methods specifically designed for RSI-SOD, such as DAFNet [35], demonstrate superior saliency prediction performance.

In geology and resource exploration, SOD supports the identification of mineral-rich areas.

Traditional SOD methods primarily utilized handcrafted features, including feature analysis and fusion [24], [25], sparse representation [26], [27], and color information [28]. Despite their foundational role, these approaches often fell short in computational efficiency and struggled to achieve the desired accuracy in practical applications, particularly when adapting to a variety of scenarios. In contrast, recent advancements in convolutional neural networks (CNNs) [29], especially those employing fully convolutional networks (FCNs) [30], have significantly propelled the NSI-SOD field forward. Innovations, such as GateNet [31], PA-KRN [32], and PGNet [33], are prime examples of such progress. Therefore, extending state-of-the-art NSI-SOD methods to the RSI-SOD domain is a natural progression. However, RSIs, captured from high altitudes by satellites and aerial sensors, exhibit characteristics distinct from natural scene images (NSIs) taken by ground-based cameras. RSIs are influenced by diverse imaging conditions, leading to significant variability in the number, orientation, and position of salient objects, often coupled with considerable scale changes. This variability challenges the efficacy of traditional central prior theories. Moreover, the texture and shape of salient objects in RSIs are typically more complex and varied. For example, elongated objects such as roads can span the entire image, against backgrounds often cluttered with interference. In addition, RSIs are subject to varying conditions such as time of capture and weather, introducing elements such as shadows, overexposure, and morphological distortions that further complicate the SOD task. This is illustrated in Fig. 1, where the performance of traditional SOD methods (VOS [34]) and mature NSI-SOD methods (GateNet [31]) on RSIs clearly demonstrates their visual limitations in handling such complex scenes.

To address these challenges, preliminary explorations focusing on RSI-SOD have emerged recently [35], [36], [37], [38], [39]. For instance, Li et al. [36] pioneered the development of the LVNet model specifically for RSI-SOD. The LVNet model employed an L-shaped dual-stream pyramid module for extracting multiscale features from images at five different resolutions. It also featured a V-shaped nested encoder–decoder module designed to highlight salient objects and suppress the background. In addition, Zhang et al. [35] introduced edge supervision and proposed DAFNet, a multitask framework where attention information is propagated from shallow to deep levels. DAFNet simultaneously infers saliency maps and edge maps, enhancing edge information. The fifth column in Fig. 1 shows the saliency prediction by DAFNet.

While existing methods have achieved notable accuracy in RSI-SOD tasks, they encounter several limitations. First, most rely on CNNs for RSI encoding, primarily capturing local image features but lacking in global contextual semantic modeling. This is a critical gap, as global semantics are essential for accurately locating salient objects, and local spatial details are vital for refining object specifics. RSIs, with their variable scales and irregular topologies, demand encoders that can simultaneously process global semantics and local spatial features. The challenge of effectively combining these aspects to address complex remote sensing scenarios remains unresolved. Second, many current approaches use static network constructions and module designs. This limits the models' ability to dynamically adjust parameters during inference, which is crucial for adapting to the varied scenes in RSIs. Third, existing methods commonly employ channel and spatial attention, which have notable drawbacks. Channel attention, in particular, tends to compress significant information, especially in deep features with numerous channels, potentially leading to the loss of crucial cues. This loss is hard to recover in the subsequent processing stages. Spatial attention, typically reliant on fixed global pooling and static convolutions, struggles to generate adaptable attention maps for the diverse scenarios encountered in RSIs. Consequently, these methods may not fully adapt to RSI characteristics, resulting in incomplete spatial structures and blurred boundaries.

To overcome these limitations, we propose a novel adaptive semantic network (ASNet) tailored for RSI-SOD. ASNet is built upon a Transformer–CNN hybrid structure and adopts a popular encoder–decoder architecture. ASNet's design encompasses several key aspects. First, to concurrently capture global semantic information and local spatial features of input images, we design the dual-branch encoder, consisting of Transformer and CNN branches. The Transformer branch is based on the pyramid vision transformer (PVT) [40], while the CNN branch leverages ResNet18 [41]. These two branches focus on global relationship modeling and complementary local fine-grained representations. Second, we propose the adaptive semantic matching module (ASMM), which dynamically generates filters to adaptively match global semantic information and local spatial information from different levels of feature representations. The matching filters are dynamically generated from three levels: scene-, region-, and position-level features. This adaptive semantic interaction helps to exploit the synergy between global and local features. Third, we enhance channel attention and spatial attention, proposing the adaptive feature enhancement module (AFEM). AFEM serves to boost the features of salient regions while suppressing redundant information, all while progressively upsampling the image. Finally, the multiscale fine-grained inference module (MFIM) is designed to tackle the detection of salient objects with complex details and shapes. Focusing on the first-level features of the CNN branch, MFIM employs multiscale dilated convolutions to capture and accentuate nuanced details. This method enhances the segmentation

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accuracy for objects across diverse scales. Guided by saliency cues, MFIM effectively generates high-quality saliency maps, significantly improving the detection and segmentation in challenging scenarios. We conduct extensive experiments on two benchmark datasets, demonstrating that ASNet outperforms existing state-of-the-art methods in RSI-SOD tasks. Furthermore, we perform multiple ablation studies to validate the rationale and effectiveness of our approach.

The core contributions are summarized as follows.

1) *ASNet:* An innovative network for RSI-SOD, adept at capturing both global and local features in RSIs and dynamically adapting to diverse remote sensing scenes.
2) *ASMM:* A novel module that adaptively generates filters, blending global and local information for enhanced feature fusion and synergy in variable RSI environments.
3) *AFEM:* An attention-based module for adaptive feature enhancement in salient regions, reducing interference and aiding in progressive image upsampling.
4) *MFIM:* Designed for multiscale detail capture, refining high-level features and generating precise RSI-SOD results, guided by saliency cues.

II. Related Work

In this section, we comprehensively review the current state of research in NSI-SOD and RSI-SOD and delve into the development of dynamic convolution technology, as detailed in Sections II-A–II-C. By exploring the primary contributions and inherent limitations of existing methods in these areas, we establish a clear motivation for our research, as outlined in Section II-D.

A. CNN-Based NSI-SOD Methods

In the NSI-SOD domain, numerous works have emerged with the primary goal of achieving pixel-level SOD using CNNs. These methods have significantly improved detection accuracy compared to traditional approaches. Some of these studies have focused on extracting and fusing multiscale features. Li and Yu [42] leveraged CNNs to extract multiscale features from images at different resolutions, leading to the learning of high-quality saliency models. Hou et al. [43] proposed an SOD method with short connections that efficiently provided rich multiscale feature maps. Hu et al. [44] optimized multilayer features through cyclic aggregation. In addition, some studies attempted to utilize edge information to guide the model in detecting salient objects with clear boundaries. Feng et al. [45] achieved precise salient object localization through feature aggregation techniques and used boundary-enhanced loss (BEL) to learn fine-grained boundaries. Sun et al. [46] embedded the U-shape edge network module in the decoder unit to integrate various features at different scales, enhancing boundary prediction performance. Furthermore, some studies introduced attention mechanisms to enable models to focus on salient features. Zhao and Wu [47] introduced the pyramid feature attention network (PFAN), which used attention mechanisms to enhance high-level contextual features and low-level spatial structural features, thereby improving saliency detection performance. Liu et al. [48] introduced pixel-level contextual attention mechanisms that selectively focused on relevant context information for each pixel, enhancing the accuracy of saliency detection. Chen et al. [49] introduced reverse attention mechanisms, enabling the model to learn missing object parts and details from predicted salient regions, thereby improving the completeness and accuracy of saliency detection. In addition, Hou et al. [43] and Wang et al. [50] applied deep supervision to enhance multiscale feature extraction capabilities.

B. CNN-Based RSI-SOD Methods

Several studies have successfully integrated the effective strategies of NSI-SOD into RSI-SOD tasks, achieving remarkable detection accuracy. Specifically, Li et al. [36] constructed the ORSSD dataset and proposed LVNet. In this model, the L-shaped dual-stream pyramid module is responsible for multiscale feature extraction, while the V-shaped nested encoder–decoder module highlights salient objects and suppresses backgrounds. Zhang et al. [35] extended the EORSSD dataset, introduced edge supervision, and proposed DAFNet. This model achieves shallow-to-deep attention information transmission, simultaneously inferring saliency maps and edge maps to enhance edge information. Cong et al. [51] introduced a multiscale attention network, designing a graph-structured relationship reasoning encoder. Huang et al. [52] fused high-level features to locate salient objects and used these high-level features to guide shallow features for refining coarse saliency maps. Li et al. [53] considered the complementarity of foreground, background, and global image-level content, integrating various information to solve the RSI-SOD problem. Tu et al. [37] proposed a joint learning model to refine both region features and edge features of salient objects simultaneously. Zhou et al. [54] introduced an edge-aware feature integration model, extracting and fusing features from three input images with different resolutions to generate saliency feature maps guided by edge information. Li et al. [55] incorporated context information from adjacent branches to assist in capturing current branch features. In addition, Gao et al. [56] attempted to use Transformer to encode images to overcome the limitations of CNN-based models. Some studies focused on low-cost weakly supervised methods. Zhang and Ma [57] proposed a weakly supervised learning method, generating pseudo-labels through Grad-CAM and progressively improving saliency maps. Ma et al. [58] achieved accurate detection and boundary localization of targets at different scales through graffiti relabeling, graffiti embedding network design, and the introduction of a multibranch edge detection network. Furthermore, some studies aimed to build lightweight models, such as FSMINet [59], CorrNet [22], and SeaNet [39].

C. Dynamic Convolution

In traditional CNNs, the kernel parameters of static convolutions are shared across all samples and positions, a static interaction approach that may lead to suboptimal feature learning. Dynamic convolution, in contrast, offers greater flexibility and adaptability. It can dynamically adjust its

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parameters based on specific input features' demands, thereby better accommodating the complexity and diversity of input data. Some studies have begun to explore the application of dynamic convolution to various tasks. In image classification, Yang et al. [60] proposed CondConv, a method that learns specialized convolutional kernels for each example, enhancing network performance and inference efficiency. Chen et al. [61] introduced dynamic convolution, aggregating multiple parallel convolution kernels dynamically based on input-dependent attention, thus increasing model complexity without expanding network depth or width. In semantic segmentation, Nirkin et al. [62] developed the real-time semantic segmentation network HyperSeg, where the encoder both encodes and generates the decoder's parameters using dynamic convolution to enhance segmentation accuracy and efficiency. Zhang et al. [63] proposed a novel decoder, NRD, for more efficient high-resolution semantic segmentation. In SOD, Pang et al. [64] explored better integration and utilization of cross-modal fusion information through dynamic filtering and a dynamic dilated pyramid module, achieving more flexible and efficient multiscale feature processing. Zhang et al. [65] developed DCFNet, a network that generates dynamic filters by extracting location-related affinities between consecutive frames, effectively enhancing SOD performance in videos. Wu et al. [66] proposed the concept of dynamic scale routing, optimizing SOD performance using dynamic pyramid convolution.

D. Motivation

In the field of NSI-SOD, strategies, such as multiscale feature extraction and fusion, edge guidance, attention mechanisms, and deep supervision, have significantly enhanced detection accuracy. However, due to the unique challenges presented by RSIs, these methods are not directly transferable to RSI-SOD tasks. In the realm of RSI-SOD, adaptations of these strategies have shown remarkable effectiveness, with models developed specifically to meet the peculiar demands of RSIs. Despite these advancements, most methods have not fully addressed the critical aspect of synergistic interaction between global and local contexts in RSI-SOD. Moreover, dynamic convolution introduces a new dimension to CNNs, offering enhanced adaptability and flexibility across various computer vision tasks. It dynamically adjusts parameters based on the specific demands of input features, providing a more targeted approach to feature learning. However, these studies have primarily focused on high-speed inference and real-time performance, not fully exploiting the potential of dynamic convolution for adaptive feature interaction.

Diverging from existing methods, our approach comprehensively encodes input images from both global and local perspectives, adaptively matching their semantic information for high-performance inference. Notably, unlike some edge-guided methods [35], [53], our approach does not require any additional auxiliary labels, thus reducing costs. Building on this, we propose a unique method that designs adaptive filtering parameters at different levels based on spatial sizes, effectively addressing the diversity and complexity of optical remote sensing scenes. We also propose a learnable gating mechanism to match different semantic features, further enhancing the adaptability of our model. This approach sets our method apart from existing solutions in the field.

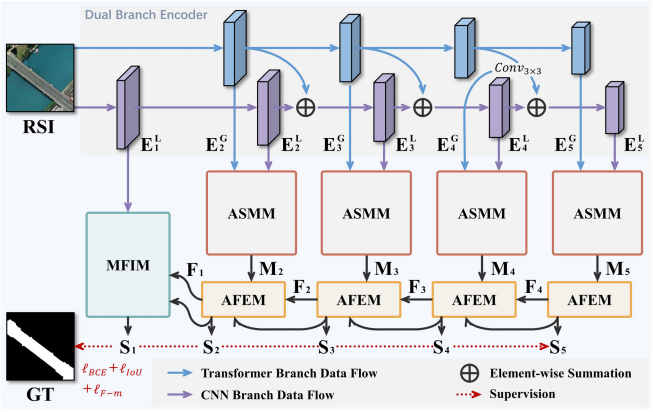


Fig. 2. Overview of the ASNet architecture, employing an encoder–decoder framework. It comprises four main components: dual-branch encoder, integrating a Transformer branch based on PVT and a CNN branch based on ResNet18 for capturing global and local features; ASMM for feature synergy; AFEM for feature enhancement and upsampling; and MFIM for refining semantic features and generating saliency maps. In addition, a comprehensive loss function based on deep supervision strategy is utilized to regulate the entire training process. Please zoom in for detailed view.

III. Method

In this section, we provide a detailed exposition of the proposed ASNet. In Section III-A, we present the overall architecture of ASNet. Subsequently, in Sections III-B–III-D, we elaborate on the three core components of ASNet: ASMM, AFEM, and MFIM, respectively. In Section III-E, we explicate the comprehensive loss function.

A. Overall Structure

The overall structure of ASNet, illustrated in Fig. 2, adopts the popular encoder–decoder architecture for SOD tasks. To begin, the dual-branch encoder is designed as a dual-branch structure to leverage the complementary features of Transformer and CNN. The Transformer branch is built upon PVT [40] (shown in blue), while the CNN branch is constructed on ResNet18 [41] (shown in purple). These branches are dedicated to capturing global contextual relationships and learning complementary local fine-grained representations. Specifically, PVT consists of four stages, each containing a patch embedding layer and a series of Transformer blocks. The image is partitioned into spatial patches of size 4×4, which, after positional encoding, are flattened and fed into the Transformer blocks. ResNet18 retains the first five feature extraction stages while discarding the fully connected layers (FCLs) used for image classification. Importantly, inputs to the third, fourth, and fifth stages of the CNN branch come not only from the preceding stage but also from the Transformer branch with the same spatial dimensions. This global semantic guidance from the Transformer branch facilitates the CNN branch in learning complementary local fine-grained features. Particularly noteworthy is the introduction of a convolutional layer in the fifth stage of the CNN branch, aimed at adjusting the channel dimensions of global features to match those

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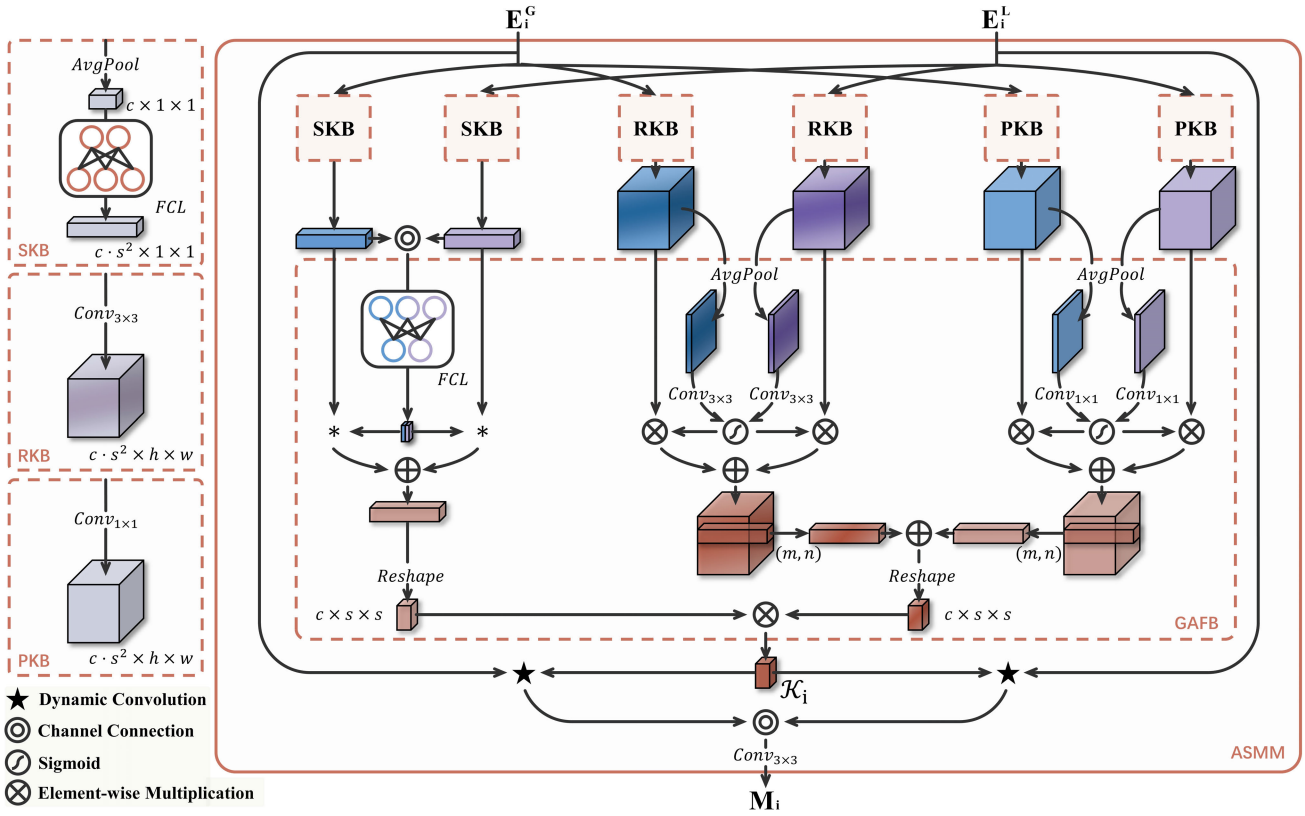


Fig. 3. Structure and workflow of the ASMM. ASMM consists of four components: SKB, RKB, PKB, and GAFB. SKB, RKB, and PKB dynamically generate kernel parameters based on overall scene characteristics, local spatial characteristics, and specific positions of input features, respectively. GAFB utilizes a gating mechanism to efficiently match global and local features. This design enables dynamic adjustment to diverse and complex remote sensing scenes, significantly enhancing adaptability and accuracy.

of local features before element-wise addition. For an input image $\boldsymbol{RSI} \in \mathbb{R}^{3\times352\times352}$, the encoded multiscale global features from the dual-branch encoder are represented as $\boldsymbol{E}_2^G \in \mathbb{R}^{64\times88\times88}$, $\boldsymbol{E}_3^G \in \mathbb{R}^{128\times44\times44}$, $\boldsymbol{E}_4^G \in \mathbb{R}^{256\times22\times22}$, and $\boldsymbol{E}_5^G \in \mathbb{R}^{512\times11\times11}$. Similarly, the multiscale local features are represented as $\boldsymbol{E}_1^L \in \mathbb{R}^{64\times176\times176}$, $\boldsymbol{E}_2^L \in \mathbb{R}^{64\times88\times88}$, $\boldsymbol{E}_3^L \in \mathbb{R}^{128\times44\times44}$, $\boldsymbol{E}_4^L \in \mathbb{R}^{256\times22\times22}$, and $\boldsymbol{E}_5^L \in \mathbb{R}^{512\times11\times11}$. These features will undergo further processing in the subsequent stages. Next, the fundamental features $\boldsymbol{E}_i^G$ and $\boldsymbol{E}_i^L$ from the four relatively higher stages ($i \in \{2, 3, 4, 5\}$) are fed into the corresponding levels of ASMM, generating semantic matching features $\boldsymbol{M}_i$ ($i \in \{2, 3, 4, 5\}$). Subsequently, AFEM utilizes $\boldsymbol{M}_i$ and the output from adjacent higher level features of AFEM as inputs, enhancing feature representations and achieving feature upsampling, resulting in features denoted as $\boldsymbol{F}_i$ ($i \in \{1, 2, 3, 4\}$). The four AFEMs are organized in a concatenated manner, progressively restoring the spatial topological structure of salient objects, thereby gradually achieving preliminary saliency predictions $\boldsymbol{S}_i$ ($i \in \{2, 3, 4, 5\}$). Finally, MFIM mines detailed information from the shallow features $\boldsymbol{E}_1^L$ as a complement to the high-level semantic features $\boldsymbol{F}_1$, guided by the saliency cues $\boldsymbol{S}_2$, to infer the final high-quality saliency map $\boldsymbol{S}_1$.

B. Adaptive Semantic Matching Module

After encoding the input image using the dual-branch encoder, two distinct sets of features, namely, global and local features, are obtained. Effectively harnessing the synergy between these features is crucial for achieving high-quality predictions. Current mainstream methods often involve direct feature concatenation or element-wise summation for aggregation. Although these methods establish information connections, their simplicity prevents them from capturing responses between different semantic features. To address this, we propose the ASMM. The objective of ASMM is to dynamically generate filters that match the semantic information between global and local features for specific samples, enabling adaptive interactions.

The structure and workflow of ASMM, illustrated in Fig. 3, comprise four core blocks: scene-aware kernel block (SKB), region-aware kernel block (RKB), position-aware kernel block (PKB), and gated adaptive filter block (GAFB). Unlike existing aggregation blocks composed of fixed-parameter convolutional layers, these blocks dynamically adjust their parameters according to changes in scenes, regions, and positions. These adjustments are achieved by SKB, RKB, and PKB, enhancing the model's adaptability to various complex remote sensing scenes. In addition, GAFB recombines kernels using a gating mechanism, generating filters that match global and local semantics. Specifically, input features $\boldsymbol{E}_i^G$ and $\boldsymbol{E}_i^L$ are convolved with filters $\mathcal{K}_i$ generated by GAFB, followed by channel-wise concatenation. Subsequently, after convolutional layer processing, channel information is merged, producing the output $\boldsymbol{M}_i$ of ASMM

$$\boldsymbol{M}_i = \mathrm{Conv}_{3\times3}\left(\left(\boldsymbol{E}_i^G \star \mathcal{K}_i\right) \circledcirc \left(\boldsymbol{E}_i^L \star \mathcal{K}_i\right)\right); \quad i \in \{2, 3, 4, 5\} \tag{1}$$

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where $\text{Conv}_{3\times3}(*)$ represents a 3×3 convolution operation with batch normalization and ReLU activation, $\star$ indicates dynamic convolution, and $\circledcirc$ signifies channel connection. Next, we will elaborate on the specific workflows of SKB, RKB, PKB, and GAFB.

1) Scene-Aware Kernel Block: It dynamically generates kernel parameters based on perceiving the overall scene characteristics of specific input features. For input features $\boldsymbol{E}_i^G$ and $\boldsymbol{E}_i^L$, global average pooling (GAP) is applied to obtain global descriptors. These descriptors are then fed into an FCL to expand their channel dimensions to s^2 times the original size, where s represents the kernel spatial size (set to 3 in this study). The resulting output vectors serve as kernel parameters $\boldsymbol{k}_i^{\text{GS}}$ and $\boldsymbol{k}_i^{\text{LS}}$, perceiving the scene characteristics

$$\boldsymbol{k}_i^{\text{GS}} = \text{FCL}\big(\text{GAP}\big(\boldsymbol{E}_i^G\big)\big); \quad \boldsymbol{k}_i^{\text{LS}} = \text{FCL}\big(\text{GAP}\big(\boldsymbol{E}_i^L\big)\big). \tag{2}$$

2) Region-Aware Kernel Block: It dynamically generates kernel parameters based on perceiving the local spatial characteristics of specific input features. Standard 3×3 convolutions are applied to input features $\boldsymbol{E}_i^G$ and $\boldsymbol{E}_i^L$, expanding their channel dimensions to s^2 times the original size, serving as the kernel parameters $\boldsymbol{k}_i^{\text{GR}}$ and $\boldsymbol{k}_i^{\text{LR}}$ for the perceived region

$$\boldsymbol{k}_i^{\text{GR}} = \text{Conv}_{3\times3}\big(\boldsymbol{E}_i^G\big); \quad \boldsymbol{k}_i^{\text{LR}} = \text{Conv}_{3\times3}\big(\boldsymbol{E}_i^L\big). \tag{3}$$

3) Position-Aware Kernel Block: It dynamically generates kernel parameters $\boldsymbol{k}_i^{\text{GP}}$ and $\boldsymbol{k}_i^{\text{LP}}$ based on perceiving specific positions of input features. Similar to RKB operations, PKB employs 1×1 convolutions to expand the channel dimensions

$$\boldsymbol{k}_i^{\text{GP}} = \text{Conv}_{1\times1}\big(\boldsymbol{E}_i^G\big); \quad \boldsymbol{k}_i^{\text{LP}} = \text{Conv}_{1\times1}\big(\boldsymbol{E}_i^L\big). \tag{4}$$

4) Gated Adaptive Filter Block: It adjusts the importance of global and local contextual information across semantics and recombines kernel parameters perceived in scenes, regions, and positions to generate adaptive filters. Specifically, kernel parameters perceived for scene characteristics in both global and local features are concatenated and fed into an FCL, compressing the channel numbers to 2 for channel-level gating

$$\alpha_1, \alpha_2 = \text{FCL}\big(\boldsymbol{k}_i^{\text{GS}} \circledcirc \boldsymbol{k}_i^{\text{LS}}\big). \tag{5}$$

The weights α_1 and α_2 act on scene kernel parameters for global and local features, respectively. The weighted sum of these two sets of features forms the scene kernel $\boldsymbol{k}_i^S$

$$\boldsymbol{k}_i^S = \alpha_1 \boldsymbol{k}_i^{\text{GS}} \oplus \alpha_2 \boldsymbol{k}_i^{\text{LS}} \tag{6}$$

where $\oplus$ denotes element-wise addition. For region-aware kernel parameters of global and local features, the GAP, 3×3 convolution, and Sigmoid activation are applied sequentially to obtain pixel-level gating mechanisms

$$\boldsymbol{\beta}_1 = \sigma\big(\text{Conv}_{3\times3}\big(\text{GAP}\big(\boldsymbol{k}_i^{\text{GR}}\big)\big)\big) \tag{7}$$

$$\boldsymbol{\beta}_2 = \sigma\big(\text{Conv}_{3\times3}\big(\text{GAP}\big(\boldsymbol{k}_i^{\text{LR}}\big)\big)\big) \tag{8}$$

where $\sigma(*)$ denotes the Sigmoid activation function. The weights $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are applied to the global and local feature's region-aware kernel parameters, respectively. The element-wise addition of these results combines them into the region kernel $\boldsymbol{k}_i^R$

$$\boldsymbol{k}_i^R = \big(\boldsymbol{\beta}_1 \otimes \boldsymbol{k}_i^{\text{GR}}\big) \oplus \big(\boldsymbol{\beta}_2 \otimes \boldsymbol{k}_i^{\text{LR}}\big) \tag{9}$$

where $\otimes$ denotes element-wise multiplication. As a result, each spatial position (m, n) can be independently extracted from $\boldsymbol{k}_i^R$, forming the convolutional kernel $\boldsymbol{k}_{i,(m,n)}^R$ at that pixel. In addition, the generation process of position-aware kernels $\boldsymbol{k}_{i,(m,n)}^P$ for global and local features is similar to that of region-aware kernels. The difference lies in using a 1×1 convolution to generate pixel-level gating, focusing only on specific positions.

Ultimately, the final filter $\mathcal{K}_i$ is obtained by reshaping and combining $\boldsymbol{k}_i^S$, $\boldsymbol{k}_{i,(m,n)}^R$, and $\boldsymbol{k}_{i,(m,n)}^P$

$$\mathcal{K}_i = \text{Reshape}\big(\boldsymbol{k}_i^S\big) \otimes \text{Reshape}\big(\boldsymbol{k}_{i,(m,n)}^R \oplus \boldsymbol{k}_{i,(m,n)}^P\big). \tag{10}$$

This process demonstrates that scene-aware dynamic kernels are applied equally to all pixels, while region and position-aware dynamic kernels are independently generated for each pixel. This design not only considers the semantic matching of global and local features but also effectively integrates scene, region, and position information, achieving adaptive feature fusion. This approach provides a more accurate and robust feature interaction method for the prediction process, enhancing the quality of predictions.

C. Adaptive Feature Enhancement Module

In the encoding process of the encoder and the feature interaction of ASMM, some disturbance factors may be introduced, affecting the accuracy of model predictions, especially in remote sensing scenes with relatively complex backgrounds. Currently, the most common methods to enhance salient features involve channel attention and spatial attention. However, these methods have limitations. Channel attention achieves enhancement by drastically compressing the number of channels, leading to severe loss of semantic information. On the other hand, spatial attention focuses only on global features and employs static convolutional operations to compute attention maps, which is not suitable for highly variable remote sensing scenes. Therefore, in Fig. 4, we propose AFEM, consisting of the channel enhancement block (CEB) and the spatial enhancement block (SEB), which comprehensively enhance salient features in both channel and spatial dimensions, suppressing interference.

1) Channel Enhancement Block: First, global max pooling (GMP) and GAP are applied to the input features to obtain global semantic representations. Subsequently, we propose a unique method to compute channel attention. Specifically, the two aforementioned channel vectors undergo semantic correlation calculations through two layers of 1-D grouped convolutions. This means that each layer of 1-D convolution achieves intergroup independence and intragroup interaction, reflecting the semantic relationships between adjacent channels. This process can be represented as

$$\begin{aligned}\boldsymbol{CA} = \sigma(&\text{GrConv}_{1\times1}(\text{GMP}(\boldsymbol{M}_i \circledcirc \boldsymbol{F}_i)) \\ &\oplus \text{GrConv}_{1\times1}(\text{GAP}(\boldsymbol{M}_i \circledcirc \boldsymbol{F}_i)))\end{aligned} \tag{11}$$

where $\boldsymbol{CA}$ represents computed channel attention and $\text{GrConv}_{1\times1}(*)$ denotes 1×1 grouped convolutions. In our experiments, setting the number of groups to 4 yielded

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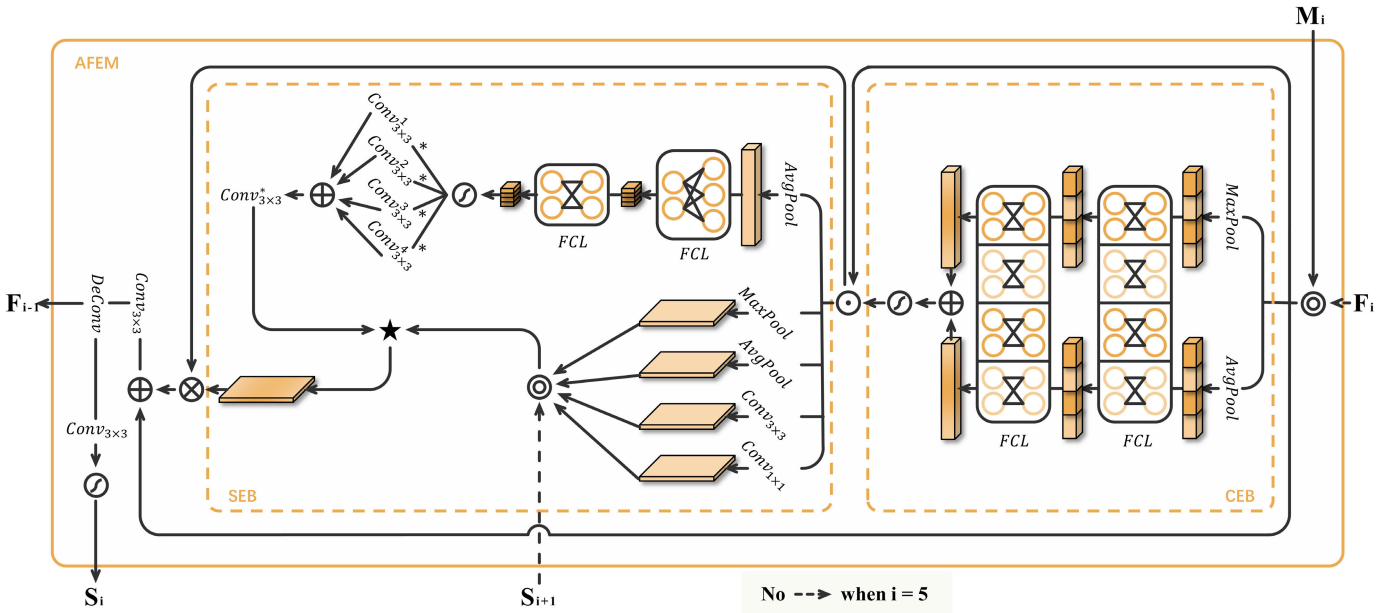


Fig. 4. Structure and workflow of the AFEM. AFEM consists of two main components: CEB and SEB. CEB enhances salient features by calculating channel attention using global pooling and 1-D grouped convolutions. SEB projects input features across multiple dimensions and integrates them with the high-level saliency map for comprehensive spatial feature enhancement. The dynamic aggregation of multiple convolution kernels based on feature attention emphasizes key areas, allowing for adaptive feature enhancement and interference suppression in salient regions.

excellent results. Finally, salient features are enhanced using channel attention

$$\boldsymbol{F}_i^{\mathrm{CA}} = \boldsymbol{CA} \odot (\boldsymbol{M}_i \circledcirc \boldsymbol{F}_i) \tag{12}$$

where $\odot$ denotes channel-wise multiplication.

2) Spatial Enhancement Block: Initially, the input features are channel-projected from global maximum, global average, local spatial, and specific positional perspectives. The projection results are then connected with the high-level saliency map $\boldsymbol{S}_{i+1}$, forming a more comprehensive spatial salient feature $\boldsymbol{F}^S$

$$\begin{aligned}\boldsymbol{F}^S = {} & \mathrm{GMP}\left(\boldsymbol{F}_i^{\mathrm{CA}}\right) \circledcirc \mathrm{GAP}\left(\boldsymbol{F}_i^{\mathrm{CA}}\right) \circledcirc \mathrm{Conv}_{3\times3}\left(\boldsymbol{F}_i^{\mathrm{CA}}\right) \\ & \circledcirc \mathrm{Conv}_{1\times1}\left(\boldsymbol{F}_i^{\mathrm{CA}}\right) \circledcirc \boldsymbol{S}_{i+1}.\end{aligned} \tag{13}$$

Next, multiple parallel convolution kernels are dynamically aggregated based on the attention of input features. Here, we employ four parallel convolution kernels. These kernels aggregate attention nonlinearly, endowing them with stronger representation capabilities

$$\gamma_k = \mathrm{FCL}\left(\mathrm{GAP}\left(\boldsymbol{F}_i^{\mathrm{CA}}\right)\right); \quad k \in \{1, 2, 3, 4\} \tag{14}$$

$$\mathbf{Conv}_{3\times3}^{*} = \sum_{k=1}^{4} \gamma_k \mathbf{Conv}_{3\times3}^{k} \tag{15}$$

where $\mathbf{Conv}_{3\times3}^{*}$ represents the dynamically generated convolutional kernel aggregated by the weights γ_k and 3×3 convolutional kernels. Finally, the dynamic convolutional kernel computes an attention map from the spatial salient feature, indicating the importance of each spatial position, selectively activating feature responses

$$\boldsymbol{SA} = \mathbf{Conv}_{3\times3}^{*} \star \boldsymbol{F}^S \tag{16}$$

$$\boldsymbol{F}_i^{\mathrm{SA}} = \boldsymbol{SA} \otimes \boldsymbol{F}_i^{\mathrm{CA}}. \tag{17}$$

This flexible and dynamic adjustment of filter parameters enables the model to fully exploit object context, adaptively enhancing significant regions' features.

The enhanced feature representation is split into two parallel branches after residual connection, 3×3 convolution, and upsampling through deconvolution. One branch is directed to the adjacent low-level side of AFEM, while the other branch infers the preliminary salient object region $\boldsymbol{S}_i$

$$\boldsymbol{F}_{i-1} = \mathrm{DeConv}\left(\mathrm{Conv}_{3\times3}\left(\boldsymbol{F}_i^{\mathrm{SA}} \oplus (\boldsymbol{M}_i \circledcirc \boldsymbol{F}_i)\right)\right) \tag{18}$$

$$\boldsymbol{S}_i = \sigma\left(\mathrm{Conv}_{3\times3}(\boldsymbol{F}_{i-1})\right). \tag{19}$$

D. Multiscale Fine-Grained Inference Module

In the wake of feature enhancement by the AFEM, the salient regions still manifest a relatively coarse nature, and the initial saliency maps may not fully reconstruct their complete spatial structures. Simultaneously, low-level features abound with rich fine-grained details, such as textures and edges, which are essential for precise saliency detection. To fully leverage these details, we have designed the MFIM, focusing particularly on extracting nuanced detail from the first-level features of the CNN branch. A key component of MFIM is the multiscale fine-grained block (MFB), which employs a series of multiscale dilated convolutions. As illustrated in Fig. 5, this approach enables MFIM to effectively capture the intricate details and textures of salient objects, thereby enhancing the adaptability of the model to the size and shape variations of salient objects. By integrating these detailed features, MFIM not only refines high-level semantic features but also aids in accurately localizing and segmenting salient objects across diverse scales and complex geometries, thus addressing a critical aspect of RSI-SOD.

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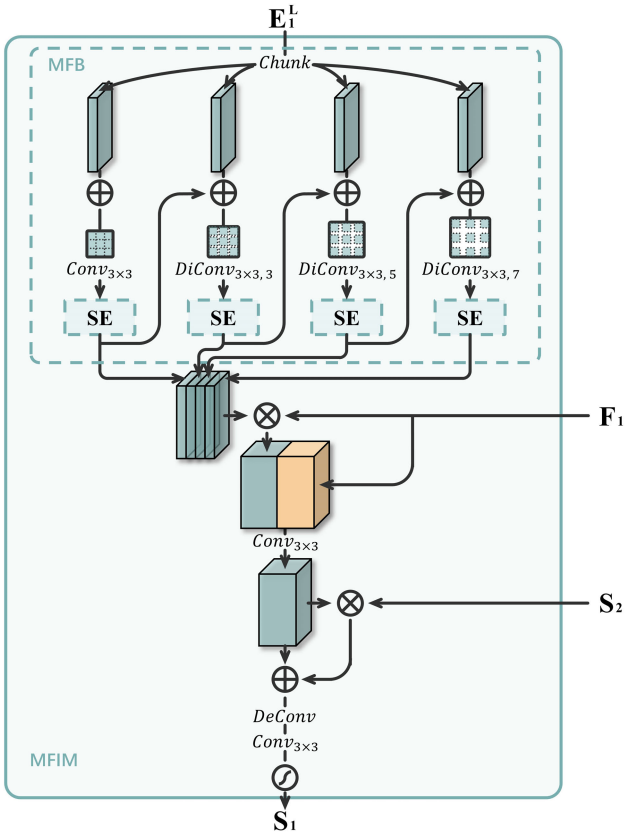


Fig. 5. Structure and workflow of the MFIM, featuring the core MFB. MFB employs multiscale dilated convolutions to capture detailed features of salient objects, accounting for their varied scales and shapes. Low-level features are divided into blocks, each processed with a distinct dilation factor and an SE block for feature refinement. Residual connections between these blocks enhance cross-scale feature interaction. Ultimately, high-level features are merged with these refined details, guided by initial saliency cues, to produce the final, accurate saliency map.

1) Multiscale Fine-Grained Block: In RSIs, salient objects exhibit significant variations in terms of position, scale, and topological structure. The MFB leverages multiscale dilated convolutions to explore intricate details. This strategy comprehensively captures multiscale information, enabling the model to adapt effectively to variations in the salient object's scale. Specifically, the low-level feature $\boldsymbol{E}_1^L$ is uniformly divided into four feature blocks $\boldsymbol{I}_p \in \mathbb{R}^{16\times256\times256}$ ($p \in \{1, 2, 3, 4\}$). Each feature block is associated with a dilated convolution with a different dilation factor and a squeeze-and-excitation (SE) block [67], where the SE block selectively filters features. In addition, to enhance cross-scale feature interactions, we introduce residual connections between the feature blocks. This process can be formulated as follows:

$$\boldsymbol{I}_p = \text{Chunk}\left(\boldsymbol{E}_1^L\right); \quad p \in \{1, 2, 3, 4\} \tag{20}$$

$$\boldsymbol{I}_p^{\text{SD}} = \begin{cases} \text{SE}\left(\text{DiConv}_{3\times3,1}\left(\boldsymbol{I}_p\right)\right); & p = 1 \\ \text{SE}\left(\text{DiConv}_{3\times3,3}\left(\boldsymbol{I}_p \oplus \boldsymbol{I}_{p-1}^{\text{SD}}\right)\right); & p = 2 \\ \text{SE}\left(\text{DiConv}_{3\times3,5}\left(\boldsymbol{I}_p \oplus \boldsymbol{I}_{p-1}^{\text{SD}}\right)\right); & p = 3 \\ \text{SE}\left(\text{DiConv}_{3\times3,7}\left(\boldsymbol{I}_p \oplus \boldsymbol{I}_{p-1}^{\text{SD}}\right)\right); & p = 4 \end{cases} \tag{21}$$

$$\boldsymbol{I}^{\text{SD}} = \boldsymbol{I}_1^{\text{SD}} \odot \boldsymbol{I}_2^{\text{SD}} \odot \boldsymbol{I}_3^{\text{SD}} \odot \boldsymbol{I}_4^{\text{SD}} \tag{22}$$

where $\boldsymbol{I}^{\text{SD}}$ represents the result after MFB processing, Chunk$(*)$ represents the chunk operation, and $\text{DiConv}_{3\times3,\ j}(*)$ represents 3×3 dilated convolution with dilation rate j.

To supplement details relevant to the salient regions, we embed the high-level features $\boldsymbol{F}_1$ into the feature $\boldsymbol{I}^{\text{SD}}$ containing rich fine-grained details to activate pertinent information and suppress irrelevant noise. Subsequently, they are fused through cascaded convolutional operations to obtain the fine-grained salient feature $\boldsymbol{I}$

$$\boldsymbol{I} = \text{Conv}_{3\times3}\left(\left(\boldsymbol{F}_1 \otimes \boldsymbol{I}^{\text{SD}}\right) \circledcirc \boldsymbol{F}_1\right). \tag{23}$$

The preliminary saliency cues $\boldsymbol{S}_2$ guide $\boldsymbol{I}$ to infer the final saliency map $\boldsymbol{S}_1$, facilitating comprehensive saliency detection

$$\boldsymbol{S}_1 = \sigma(\text{Conv}_{3\times3}(\text{DeConv}((\boldsymbol{S}_2 \otimes \boldsymbol{I}) \oplus \boldsymbol{I}))). \tag{24}$$

E. Comprehensive Loss

Several research methods [54], [68] have verified the stability and improvement in detection accuracy achieved by employing mixed and complementary loss functions during the training process. Based on a deep supervision strategy, we have defined a comprehensive loss function to constrain the entire training process, as illustrated in Fig. 2. During the training stage, we applied pixel-level BCE loss (ℓ_{BCE}), map-level intersection over union (IoU) loss (ℓ_{IoU}), and F-m loss (ℓ_{F-m}) [69] to the saliency map $\boldsymbol{S}_i$. The overall loss $\mathcal{L}$ is represented as follows:

$$\mathcal{L} = \sum_{i=1}^{5} \ell_{\text{BCE}}(\text{Up}(\boldsymbol{S}_i), \boldsymbol{G}) + \ell_{\text{IoU}}(\text{Up}(\boldsymbol{S}_i), \boldsymbol{G}) + \ell_{F-m}(\text{Up}(\boldsymbol{S}_i), \boldsymbol{G}) \tag{25}$$

where Up$(*)$ denotes bilinear upsampling with different scale factors and $\boldsymbol{G}$ denotes the ground truth.

IV. Experiments

In this section, we first introduced the experimental procedures. Subsequently, experiments were conducted on two benchmark datasets, and comprehensive comparisons were made with competitive methods. Furthermore, we demonstrated the rationality and effectiveness of our approach through ablation experiments. Finally, a detailed discussion was presented.

A. Experimental Protocol

1) Datasets: To facilitate comparison with other methods, we utilized two existing benchmark datasets for RSI-SOD in our experiments. The ORSSD dataset, provided by Li et al. [36], was collected from Google Earth and other RSI datasets, making it the first publicly available challenging RSI-SOD dataset. It comprises 800 RSIs, with 600 images designated for training and 200 for testing, each accompanied by corresponding pixel-level annotations. The EORSSD dataset, developed as an extension to ORSSD by Zhang et al. [35], enriches the collection with additional images that exhibit semantic complexity and challenges. It encompasses 2000 RSIs, with 1400 images allocated for training and 600 for testing. Both datasets are widely adopted in the field, enabling a fair and comprehensive evaluation of our proposed method.

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TABLE I

QUANTITATIVE COMPARISON RESULTS ON THE EORSSD AND ORSSD DATASETS. THE SYMBOLS ↑ AND ↓ INDICATE THAT HIGHER AND LOWER VALUES, RESPECTIVELY, ARE PREFERABLE FOR EACH METRIC. (T) DENOTES THAT THE ENCODER EMPLOYS A TRANSFORMER BACKBONE. THE TOP THREE RESULTS ARE HIGHLIGHTED IN **RED**, **BLUE**, AND **GREEN** FOR CLEAR DISTINCTION

Methods	Params (M)	FLOPs (G)	Speed (FPS)	EORSSD								ORSSD							
				$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$S_\alpha\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$S_\alpha\uparrow$	$M\downarrow$
Traditional NSI-SOD																			
DSG	-	-	-	.5232	.4597	.4012	.7260	.6594	.6188	.6420	.1246	.6238	.5747	.5657	.7912	.7337	.7532	.7195	.1041
SMD	-	-	-	.5884	.5473	.4081	.7697	.7286	.6416	.7101	.0771	.6692	.6214	.5568	.8230	.7745	.7682	.7640	.0715
RCRR	-	-	-	.3995	.3685	.3347	.6882	.5946	.5636	.6007	.1644	.5591	.5126	.4876	.7651	.7021	.6950	.6849	.1277
Traditional RSI-SOD																			
VOS	-	-	-	.2765	.2107	.1836	.5982	.4886	.4767	.5082	.2096	.3471	.2717	.2633	.6514	.5352	.5826	.5366	.2151
CMC	-	-	-	.3268	.2692	.2007	.6803	.5894	.4890	.5798	.1057	.3913	.3454	.3108	.7064	.6417	.5996	.6033	.1267
SMFF	-	-	-	.5176	.2992	.2083	.7744	.5197	.5014	.5401	.1434	.4417	.2684	.2496	.7402	.4920	.5676	.5312	.1854
CNN-based NSI-SOD																			
EGNet	108.07	291.9	9	.7880	.6967	.5379	.9570	.8775	.7566	.8601	.0110	.8332	.7500	.6452	.9731	.9013	.8226	.8721	.0216
PoolNet	53.63	123.4	25	.7545	.6406	.4611	.9292	.8193	.6836	.8207	.0210	.7706	.6999	.6166	.9343	.8650	.8124	.8403	.0358
CPDNet	47.85	9.4	84	.8093	.7660	.6638	.9390	.8977	.8663	.8874	.0111	.8524	.8243	.7717	.9438	.9207	.9211	.8954	.0187
GCPA	67.06	54.3	23	.8347	.7905	.6723	.9524	.9167	.8647	.8869	.0102	.8687	.8433	.7861	.9509	.9341	.9205	.9026	.0168
MINet	47.56	146.3	12	.8344	.8174	.7705	.9442	.9346	.9243	.9040	.0093	.8761	.8574	.8251	.9545	.9454	.9423	.9040	.0144
GateNet	100.02	108.3	25	.8566	.8228	.7109	.9610	.9385	.8909	.9114	.0095	.8871	.8679	.8229	.9664	.9538	.9428	.9186	.0137
U2Net	44.01	376.2	25	.8732	.8329	.7221	.9649	.9373	.8989	.9199	.0076	.8738	.8492	.8038	.9539	.9387	.9326	.9162	.0166
SUCA	117.71	56.4	24	.8229	.7949	.7260	.9520	.9277	.9082	.8988	.0097	.8484	.8237	.7748	.9584	.9400	.9194	.8989	.0145
PA-KRN	141.06	617.7	16	.8639	.8358	.7993	.9616	.9536	.9416	.9192	.0104	.8890	.8727	.8548	.9680	.9620	.9579	.9239	.0139
PFSNet	31.2	73.6	67	.8705	.8583	**.8396**	.9659	.9610	.9342	.9233	.0079	.8832	.8964	.8716	.9604	.9528	.9398	.9187	.0149
PGNet	73.71	77.4	43	.7887	.7775	.7402	.9315	.9178	.8858	.8757	.0153	.8338	.8192	.7921	.9420	.9218	.9105	.8894	.0225
ICON$_{(T)}$	92.10	52.6	37	.8622	.8371	.8065	.9687	.9619	.9497	.9185	.0073	.8939	.8671	.8444	.9704	.9637	.9554	.9256	.0116
CNN-based RSI-SOD																			
LVNet	-	-	1.4	.7794	.7328	.6284	.9254	.8801	.8445	.8630	.0146	.8263	.7995	.7506	.9456	.9259	.9195	.8815	.0207
DAFNet	29.35	68.5	26	.8614	.7845	.6427	**.9861**	.9291	.8446	.9166	**.0060**	.8928	.8511	.7876	.9771	.9539	.9360	.9191	.0113
SARNet	25.91	129.7	47	.8719	.8541	.8304	.9620	.9555	.9536	.9240	.0099	.8850	.8619	.8512	.9557	.9477	.9464	.9134	.0187
MJRBM	43.54	95.7	32	.8289	.7864	.6967	.9461	.9075	.8947	.8971	.0112	.8854	.8557	.7982	.9633	.9478	.9410	.9182	.0138
EMFINet	107.26	480.9	25	.8720	.8486	.7984	.9711	.9604	.9501	.9290	.0084	.9002	.8856	.8617	.9737	.9671	.9663	.9366	.0109
MCCNet	67.65	112.8	95	**.8904**	.8604	.8137	**.9755**	**.9685**	.9538	**.9327**	**.0066**	**.9155**	**.9054**	**.8957**	**.9800**	**.9758**	**.9735**	**.9437**	**.0087**
FSMINet	3.56	9.7	54	.7881	.7658	.7165	.9281	.9185	.9038	.8774	.0136	.7804	.7613	.7351	.9007	.8930	.8975	.8343	.0490
RRNet	68.98	172.3	109	.8870	.8431	.8024	.9608	.9309	.8977	.9181	.0085	.9088	.8827	.8549	.9654	.9474	.9368	.9264	.0129
ERPNet	56.48	87.2	50	.8598	.8301	.7453	.9589	.9399	.9211	.9131	.0097	.8791	.8523	.8137	.9530	.9321	.9411	.8655	.0379
CorrNet	4.09	21.1	100	.8778	**.8620**	.8311	.9696	.9646	**.9593**	.9289	.0083	.9129	**.9002**	**.8875**	.9790	.9746	.9721	.9380	.0098
ACCoNet	102.55	180.0	81	.8837	.8552	.7969	.9727	.9653	.9450	.9290	.0074	**.9149**	.8971	.8806	**.9796**	**.9754**	.9721	**.9437**	**.0088**
HFANet$_{(T)}$	60.53	68.3	26	**.8876**	**.8681**	**.8365**	.9740	**.9679**	**.9644**	**.9380**	.0070	.9112	.8981	.8819	.9770	.9712	**.9722**	**.9399**	.0092
Ours	65.43	20.1	61	**.8959**	**.8770**	**.8672**	**.9783**	**.9745**	**.9748**	**.9345**	**.0055**	**.9172**	**.9072**	**.8986**	**.9803**	**.9764**	**.9795**	**.9441**	**.0081**

2) Implementation Details: All experiments were conducted on a platform equipped with NVIDIA GeForce GTX 3090 GPU using the PyTorch framework. Each RSI was resized to 352 × 352. Data augmentation techniques, including flipping and rotation, were applied to augment the training set, resulting in seven additional samples per image. We trained our model for 60 epochs on both ORSSD and EORSSD datasets. The initial learning rate was set to $1e^{-4}$ and decayed to $1e^{-5}$ after 45 iterations. The batch size was set to 8, and the Adam optimization strategy was employed.

3) Evaluation Metrics: We employed the following objective evaluation metrics to compare our method with others. F-measure (F_β) [70] comprehensively evaluates the quality of saliency maps by harmonizing precision and recall. We utilized its maximum, mean, and adaptive values ($F_\beta^{\max}$, F_β^{mean}, and F_β^{adp}) for a comprehensive assessment, where β represents the weight of precision, with $\beta^2 = 0.3$. E-measure (E_ξ) [71] assesses saliency maps based on the difference between local pixel information and global mean information. We similarly utilized its maximum, mean, and adaptive values ($E_\xi^{\max}$, E_ξ^{mean}, and E_ξ^{adp}). S-measure (S_α) [72] evaluates the structural similarity between saliency maps and ground truth maps from both object and region perspectives. The parameter α balances the weight between object score and region score, set to $\alpha = 0.5$. Mean absolute error (M) [73] assesses the mean absolute error between saliency maps and ground truth maps pixel-wise. Precision-recall curve (*PR*) and F-measure curve were also considered. Among these metrics, smaller M values are preferable, while larger values are desired for other metrics.

B. Comparison With State-of-the-Arts

We conducted a comprehensive comparison of our proposed method with 30 highly representative and competitive SOD techniques. All comparative algorithms' saliency maps were either publicly available or generated from their source code. These methods can be categorized into four groups.

1) *Traditional NSI-SOD:* DSG [27], SMD [74], and RCRR [75].
2) *Traditional RSI-SOD:* VOS [34], CMC [76], and SMFF [25].
3) *CNN-Based NSI-SOD:* EGNet [77], PoolNet [78], CPDNet [79], GCPA [80], MINet [81], GateNet [31], U2Net [82], SUCA [83], PA-KRN [32], PFSNet [84], PGNet [33], and ICON [85].
4) *CNN-Based RSI-SOD:* LVNet [36], DAFNet [35], SARNet [52], MJRBM [37], EMFINet [86], MCCNet [53], FSMINet [59], RRNet [51], ERPNet [38], CorrNet [39], ACCoNet [55], and HFANet [87].

1) Quantitative Comparison: Table I presents the quantitative comparison results of our ASNet method with these competitive methods on EORSSD and ORSSD. Overall, CNN-based SOD methods outperform traditional handcrafted

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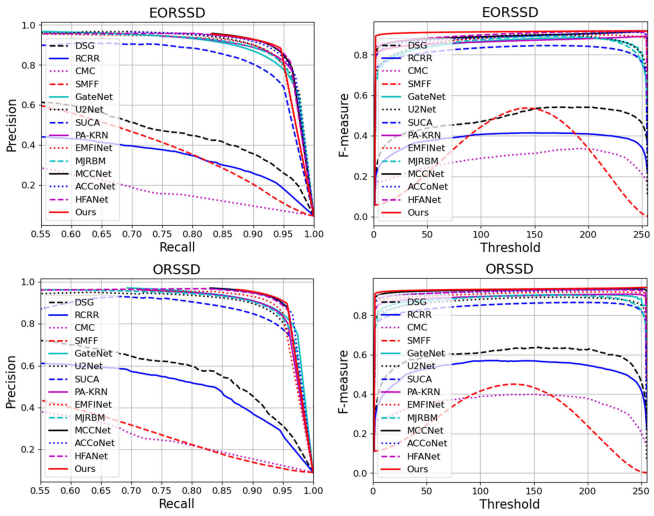


Fig. 6. *PR* and F-measure curves for our method and other representative methods on the EORSSD and ORSSD datasets. Please zoom in for detailed view.

methods significantly. RSI-SOD methods based on CNNs perform better than NSI-SOD methods, emphasizing the necessity of specialized methods for addressing RSI-SOD challenges. Specifically, on EORSSD, our method ranks first in six metrics and second in two metrics, surpassing other methods comprehensively. Although slightly inferior to DAFNet and HFANet in $E_\xi^{\max}$ and S_α, respectively, the difference is marginal, while our method outperforms them significantly in other metrics. On ORSSD, our method achieves the best performance across all metrics. Notably, unlike our ASNet, methods, such as DAFNet and MCCNet, require edge information as auxiliary supervision during training, incurring additional costs. In addition, the *PR* and F-measure curves in Fig. 6 demonstrate the absolute dominance of our method and are consistent with the aforementioned metric evaluations.

2) Qualitative Comparison: Fig. 7 provides a visual comparison between our method and other competitive methods in various challenging scenarios.

1) *Salient Objects in Large Regions:* RSI-SOD methods capture salient objects in these three scenarios, but most detected salient objects are incomplete and structurally ambiguous. In contrast, our method highlights the entire region comprehensively and maintains clear object boundaries.
2) *Multiple Tiny Salient Objects:* In the first two scenarios, some methods miss salient objects to varying degrees. In the third scenario, almost all methods are confused by obstacles, erroneously detecting more nonsalient areas. Clearly, our method excludes interference, accurately locates all tiny salient objects, and outlines their edges clearly, demonstrating strong scale adaptability.
3) *Salient Objects With Interference:* In the first scenario, interfering buildings attract the attention of many methods, leading to erroneous detection of more areas. Moreover, the second scenario renders all representative methods ineffective, generating defective saliency maps. In contrast, our method resists interference, effectively highlighting salient objects without any redundant areas.
4) *Narrow Extended Salient Objects:* In the first scenario, the failure of the vast majority of the methods to highlight the entire river, demonstrating strong inadequacy. Conversely, our method accurately delineates narrow, irregularly structured salient objects. Worth noting is that all methods overlook the side branches of the salient objects in the second scenario, and most methods consider the salient regions in the third scenario as background. In comparison, our method still achieves promising results, capturing these salient cues and achieving a breakthrough from 0 to 1.

3) Complexity Comparison: The network parameters (Params), floating-point operations (FLOPs), and inference speed (Speed) in Table I reflect the model complexity of different methods. Clearly, our model is at a leading level. Specifically, our ASNet comprises 65.43M parameters, 20.1G FLOPs, and achieves 61 frames per second (FPS). The backbone network accounts for approximately 64.45M parameters, while the key modules, ASMM, AFEM, and MFIM, account for about 0.19M, 0.03M, and 0.10M parameters, respectively. From a qualitative analysis, to simultaneously model global and local contexts, we designed a dual-branch encoder, concentrating the model parameters mainly in the backbone network. Despite this, due to the lightweight design concepts employed in our key modules, such as adaptive filtering, chunked convolution, and dilated convolution, our model maintains relatively low complexity. In comparison to FSMINet, which focuses on lightweight methods, their complexity is lower than ours. However, it is crucial to note that their detection performance lags significantly behind ours. In summary, our model strikes a fine balance between performance and computational complexity, being effective and efficient, making it suitable for practical application scenarios.

4) Performance and Adaptability Analysis: The exceptional performance of ASNet is attributed to its unique network architecture that skillfully integrates the strengths of Transformer and CNN, thereby achieving a profound fusion of global semantic information and local spatial features. ASNet's dual-branch encoder not only bolsters global relationship modeling capabilities but also captures intricate local features with precision. In addition, the ASMM, through its dynamically generated filters, adaptively matches features across different levels, a crucial factor in handling salient objects with varying scales and complex shapes. For instance, in scenarios featuring large salient objects or multiple tiny targets, ASNet accurately locates and clearly outlines the object boundaries, demonstrating its adaptability across different scales. Moreover, the adaptive feature enhancement capabilities of AFEM further improve the model's accuracy in recognizing salient regions while effectively suppressing background noise. In complex backgrounds and interference conditions, AFEM excels in reducing distractions from nonsalient areas and highlighting focal regions. Furthermore, the MFIM, by mining multiscale detailed information from lower level features, provides significant support for refining high-level semantic features, playing a key role in accurately segmenting salient objects. In various remote sensing scenarios, ASNet exhibits commendable robustness. It adeptly adapts to images with

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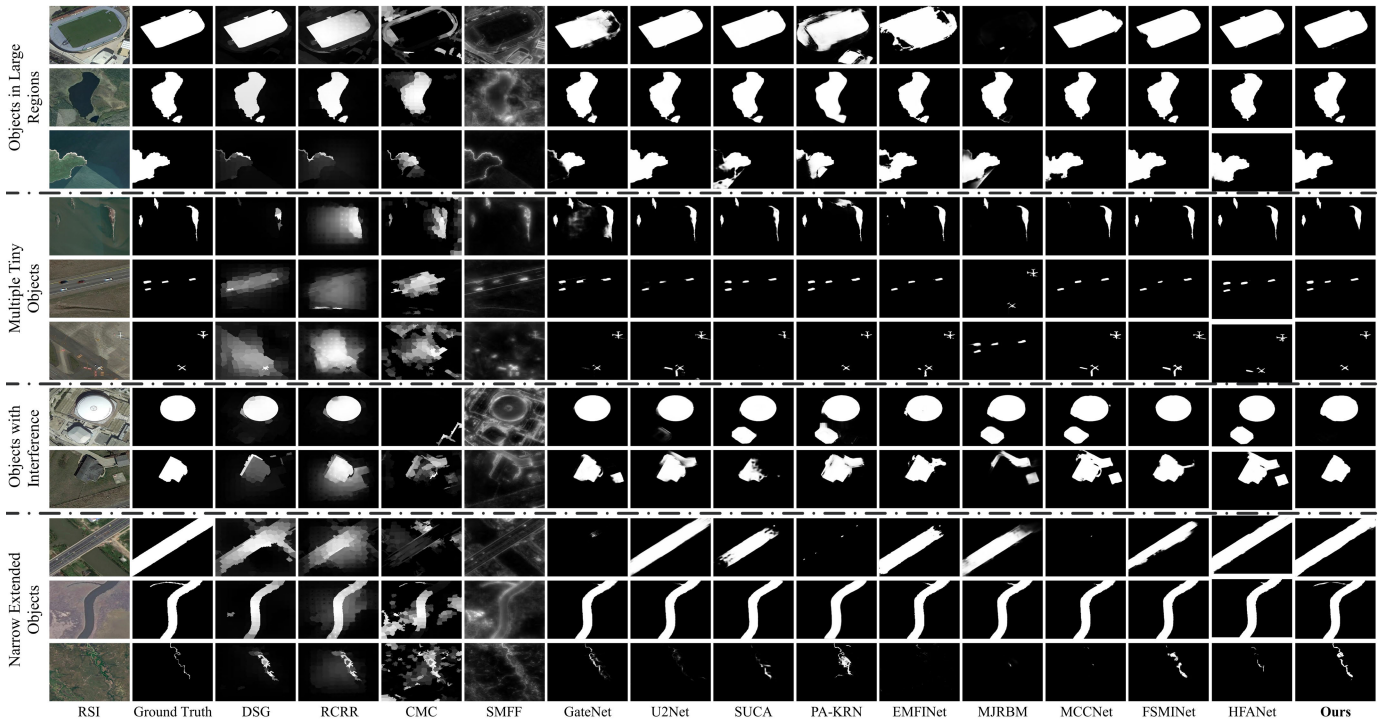


Fig. 7. Qualitative comparisons of our method with other representative approaches in four challenging scenarios. Please zoom in for detailed view.

significant changes in lighting conditions or complex terrain features, delivering accurate saliency predictions. This aspect was distinctly evident in our experimental results, especially in processing challenging remote sensing scenes, where ASNet's stability and adaptability clearly surpassed other methods.

Overall, ASNet excels not only in performance but also in adaptability and robustness across a range of remote sensing scenarios. Its innovative network design and module innovation provide an effective solution for RSI-SOD, laying a solid foundation for future research in related fields.

TABLE II

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE CONTRIBUTION OF EACH MODULE IN ASNET. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
Baseline (T)	.8652	.9556	.0092	.8861	.9593	.0125
Baseline (C)	.8639	.9544	.0095	.8833	.9570	.0137
Baseline (T+C)	.8669	.9570	.0090	.8870	.9611	.0117
Baseline (T+C)+ASMM	.8854	.9733	.0067	.9083	.9767	.0092
Baseline (T+C)+AFEM	.8789	.9688	.0079	.9011	.9718	.0101
Baseline (T+C)+MFIM	.8718	.9629	.0086	.8899	.9637	.0110
ASNet	**.8959**	**.9783**	**.0055**	**.9172**	**.9803**	**.0081**

C. Ablation Studies

To validate the effectiveness and rationality of our model design, we conducted comprehensive ablation experiments on the EORSSD and ORSSD datasets.

1) Contribution of Each Module in ASNet: In this experiment, six variants were designed: 1) Baseline (T), utilizing a U-shaped network architecture with only the Transformer branch encoder; 2) Baseline (C), employing a U-shaped network architecture with only the CNN branch encoder; 3) Baseline (T+C), integrating both Transformer and CNN branches using the dual-branch encoder; 4) Baseline (T+C)+ASMM; 5) Baseline (T+C)+AFEM; and 6) Baseline (T+C)+MFIM.

The quantitative results in Table II demonstrate that on the EORSSD dataset, Baseline (T) achieved performance of 86.52% on $F_\beta^{\max}$, 95.56% on $E_\xi^{\max}$, and 0.92% on M. Baseline (C) achieved 86.39% on $F_\beta^{\max}$, 95.44% on $E_\xi^{\max}$, and 0.95% on M. When the dual-branch encoder was employed, Baseline (T+C) reached 86.69% on $F_\beta^{\max}$, 95.70% on $E_\xi^{\max}$, and 0.90% on M. With the incorporation of different modules into the Baseline (T+C), the model's performance improved. Specifically, Baseline (T+C)+ASMM showed improvements of 1.85%, 1.63%, and 0.23% on each of the three metrics. Baseline (T+C)+AFEM improved by 1.20%, 1.18%, and 0.11%. Baseline (T+C)+MFIM showed enhancements of 0.49%, 0.59%, and 0.04%. The complete ASNet achieved the best performance, reaching 89.59% on $F_\beta^{\max}$, 97.83% on $E_\xi^{\max}$, and 0.55% on M. Similarly, on the ORSSD dataset, ASNet achieved 91.72% on $F_\beta^{\max}$, 98.03% on $E_\xi^{\max}$, and 0.81% on M, outperforming other variants. Thus, each module in ASNet significantly contributed to the model's final performance.

Visualization in Fig. 8 displayed heatmaps of different variants. It is observed that while each variant had its imperfections, dual-branch encoder and ASMM effectively adapted to optical remote sensing scenes, capturing the overall structure of salient objects. AFEM aided in enhancing salient regions and suppressing the background, while MFIM refined the edges of salient objects. Only when these modules synergized could the complete ASNet generate accurate and sharply delineated saliency maps.

2) Effectiveness of ASMM: In this experiment, four variants of ASMM were designed: 1) ASMM w/o SKB, removing SKB

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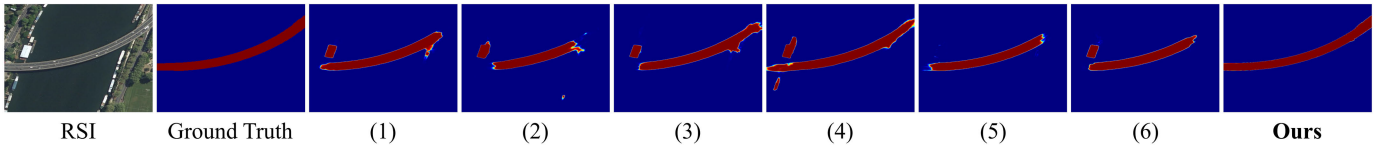


Fig. 8. Qualitative results of ablation experiments on the contribution of each module in ASNet. (1) Baseline (T). (2) Baseline (C). (3) Baseline (T+C). (4) Baseline (T+C)+ASMM. (5) Baseline (T+C)+AFEM. (6) Baseline (T+C)+MFIM. Please zoom in for detailed view.

TABLE III

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE EFFECTIVENESS OF ASMM. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
ASNet (ASMM)	**.8959**	**.9783**	**.0055**	**.9172**	**.9803**	**.0081**
ASMM w/o SKB	.8861	.9714	.0066	.9041	.9714	.0103
ASMM w/o RKB	.8876	.9723	.0062	.9074	.9751	.0097
ASMM w/o PKB	.8899	.9740	.0060	.9090	.9769	.0091
ASMM w/o GAFB	.8855	.9710	.0068	.9029	.9705	.0109

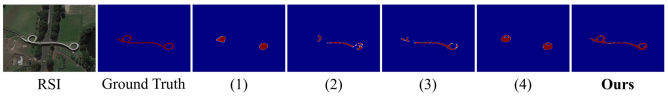


Fig. 9. Qualitative results of ablation experiments on the Effectiveness of ASMM. (1) ASMM w/o SKB. (2) ASMM w/o RKB. (3) ASMM w/o PKB. (4) ASMM w/o GAFB. Please zoom in for detailed view.

TABLE IV

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE EFFECTIVENESS OF AFEM. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
ASNet (AFEM)	**.8959**	**.9783**	**.0055**	**.9172**	**.9803**	**.0081**
AFEM w/o CEB	.8812	.9700	.0077	.8977	.9698	.0090
AFEM w/o SEB	.8824	.9709	.0073	.8997	.9723	.0092

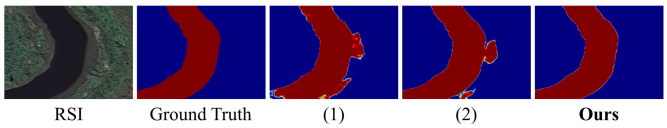


Fig. 10. Qualitative results of ablation experiments on the Effectiveness of AFEM. (1) AFEM w/o CEB. (2) AFEM w/o SEB. Please zoom in for detailed view.

TABLE V

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE EFFECTIVENESS OF MFIM. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
ASNet (MFIM)	**.8959**	**.9783**	**.0055**	**.9172**	**.9803**	**.0081**
MFIM w/ $\text{Conv}_{3\times3}$	.8922	.9747	.0058	.9129	.9750	.0088
MFIM w/o SE	.8899	.9767	.0060	.9114	.9741	.0092
MFIM w/o RC	.8936	.9743	.0057	.9138	.9757	.0086

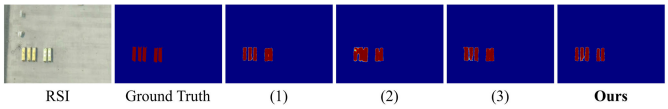


Fig. 11. Qualitative results of ablation experiments on the effectiveness of MFIM. (1) MFIM w/ $\text{Conv}_{3\times3}$. (2) MFIM w/o SE. (3) MFIM w/o RC. Please zoom in for detailed view.

from ASMM; 2) ASMM w/o RKB; 3) ASMM w/o PKB; and 4) ASMM w/o GAFB.

The quantitative comparisons in Table III indicated that on the EORSSD dataset, these ASMM variants exhibited varying degrees of performance degradation compared to ASNet. Specifically, ASMM w/o SKB decreased by 0.98%, 0.69%, and 0.11% on $F_\beta^{\max}$, $E_\xi^{\max}$, and M, respectively. ASMM w/o RKB decreased by 0.83%, 0.60%, and 0.07% on $F_\beta^{\max}$, $E_\xi^{\max}$, and M, respectively. ASMM w/o PKB decreased by 0.60%, 0.43%, and 0.05% on $F_\beta^{\max}$, $E_\xi^{\max}$, and M, respectively. ASMM w/o GAFB, due to its inability to adaptively harmonize different levels of dynamic kernels, decreased by 1.04%, 0.73%, and 0.13% on $F_\beta^{\max}$, $E_\xi^{\max}$, and M, respectively. Similarly, the same trend was observed on the ORSSD dataset.

Visual inspection in Fig. 9 highlighted the shortcomings of ASMM variants. ASMM w/o SKB struggled to reconstruct the overall structure of salient objects globally. ASMM w/o RKB failed to capture local details of salient objects. ASMM w/o PKB performed poorly in handling fine details of salient objects. ASMM w/o GAFB, due to its inability to adapt to different levels of dynamic kernels, exhibited strong maladaptation, struggling to meet scene requirements. In contrast, the saliency maps generated by the complete ASMM structure closely resembled ground truth, demonstrating superior overall performance.

3) Effectiveness of AFEM: In this experiment, two variants of AFEM were designed: 1) AFEM w/o CEB, removing CEB from AFEM, and 2) AFEM w/o SEB.

Quantitative comparisons in Table IV across both datasets revealed that AFEM w/o CEB and AFEM w/o SEB did not perform as well as ASNet, underscoring the effectiveness of AFEM. Moreover, visual results in Fig. 10 highlighted that both AFEM w/o CEB and AFEM w/o SEB failed to achieve the desired results. Only the complete AFEM structure effectively eliminated background noise and emphasized salient regions.

4) Effectiveness of MFIM: In this experiment, three variants of MFIM were designed: 1) MFIM w/ $\text{Conv}_{3\times3}$, replacing dilated convolutions in MFIM with regular 3×3 convolutions; 2) MFIM w/o SE, removing the SE from MFIM; and 3) MFIM w/o RC, removing residual connections between blocks in MFIM.

The quantitative results in Table V and the heatmap in Fig. 11 demonstrated the rationality of MFIM's design. Effective strategies, such as multiscale dilated convolutions, feature filtering, and residual connections, collectively promoted the refinement of details in salient objects.

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TABLE VI

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE DYNAMIC ADAPTABILITY OF ASNET. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
ASNet	**.8959**	**.9783**	**.0055**	**.9172**	**.9803**	**.0081**
ASNet w/o DC	.8783	.9685	.0082	.9007	.9714	.0106

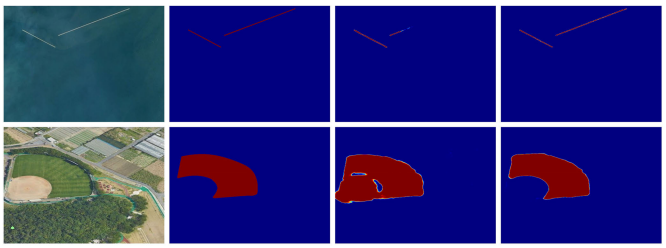


Fig. 12. Qualitative results of ablation experiments on the dynamic adaptability of ASNet. (1) ASNet w/o DC. Please zoom in for detailed view.

5) Dynamic Adaptability of ASNet: In this experiment, we designed a specific variant.

ASNet w/o DC: In this variant, all dynamic convolutions in ASNet were replaced with standard 3×3 convolutions, and the CEB and SEB components within the AFEM were substituted with the popular convolutional block attention module (CBAM) [88].

The quantitative results of this experiment are presented in Table VI, while the qualitative analysis is illustrated in Fig. 12. These results affirm the dynamic adaptability of ASNet. Compared to ASNet w/o DC, ASNet demonstrated significant improvements across three key performance metrics. Moreover, visual comparisons across diverse scenes revealed that ASNet more accurately identifies and segments salient objects, especially in complex and varied remote sensing scenarios. These experimental outcomes not only validate the efficient dynamic adaptability of ASNet in processing diverse optical remote sensing scenes but also highlight the critical role of its dynamic convolution module in adapting to the varying RSI characteristics.

6) Flexibility of ASNet: In this experiment, a variant was provided.

ASNet (PVTv2-ResNet50): We replaced the PVT and ResNet18 in ASNet's backbone (dual-branch encoder) with PVTv2 and ResNet50.

Performance comparisons in Table VII revealed that ASNet (PVTv2-ResNet50), utilizing a more powerful encoder backbone, exhibited improvements in all evaluation metrics compared to our original method. However, this enhancement came at the cost of increased network parameters and computational complexity. Therefore, the choice of using PVT and ResNet18 to construct the dual-branch encoder was a balanced decision between performance and complexity. In addition, our method exhibited strong adaptability to different backbones, showcasing low coupling in its core modules and ease of extension to other Transformer and CNN encoder networks. These results comprehensively validated

TABLE VII

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE FLEXIBILITY OF ASNET. THE BEST RESULT IS BOLDED

Models	EORSSD			ORSSD		
	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$M\downarrow$
ASNet	.8959	.9783	.0055	.9172	.9803	.0081
ASNet (PVTv2-ResNet50)	**.8997**	**.9852**	**.0051**	**.9286**	**.9884**	**.0069**

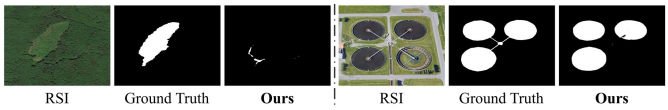


Fig. 13. Visualization of two failure cases predicted by our method. Please zoom in for detailed view.

the outstanding performance and flexibility of our proposed ASNet model, providing an effective solution for RSI-SOD.

D. Discussion

We delve into an in-depth discussion of the potential limitations of ASNet and propose directions for future work. By analyzing failure cases and identifying current method constraints, our aim is to provide valuable insights for future research in this domain.

1) Failure Cases and Potential Limitations: Despite ASNet's impressive performance across various standard benchmarks, it exhibits certain limitations in extreme and complex scenarios. This is exemplified in Fig. 13, which presents two notable failure cases.

In the first case, the salient object is an island with blurred boundaries and coloration very similar to the background. ASNet successfully highlights some distinctive edge cues but fails to comprehensively identify the entire island region. This suggests a limitation in dealing with objects that closely resemble their surroundings in terms of texture and color, a common challenge in RSIs. The second case illustrates an issue where ASNet misinterprets the connecting parts of salient regions as background. This misinterpretation arises due to the absence of additional boundary labels in the training process, leading to a lack of focus on subtle edges. This underlines the need for enhanced edge detection capabilities within the model.

These examples highlight the challenges in handling objects with low contrast against their background and the necessity to refine edge detection mechanisms in our model.

2) Future Work: Reflecting on the limitations identified, our future research will be oriented in three key directions. First, we plan to enhance edge detection capabilities by developing methods for capturing reliable edge information of salient objects directly from raw images, with the aim to improve the accuracy of boundary localization in challenging scenarios. This is expected to refine the precision of saliency detection significantly. Second, we are committed to the exploration of unsupervised RSI-SOD techniques, motivated by recent advancements in unsupervised learning approaches [89]. Our goal is to learn saliency features without the reliance on annotated data, potentially reducing training costs and efforts significantly. Third, considering the practical application

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demands, especially in mobile and on-device deployments, we acknowledge the necessity for high-performance yet lightweight models. The current tendency toward increased computational complexity is impractical for real-world applications, prompting us to explore more efficient network architectures and optimization techniques to balance performance and computational overhead.

Through these future research directions, our aim is to overcome the existing limitations of ASNet and further advance the field of RSI-SOD. We hope that these efforts will lead to a more comprehensive and widely applicable solution for a variety of practical application scenarios.

V. Conclusion

In this article, we propose a novel ASNet based on Transformer–CNN, providing an innovative perspective for the solution and network design of RSI-SOD. Specifically, through our designed dual-branch encoder and ASMM, ASNet simultaneously models global relationships and learns complementary local fine-grained representations of input images. It adaptively matches semantic information based on specific samples, triggering synergistic effects to adapt to significant variations in salient object properties within optical remote sensing scenes. Furthermore, to handle the potential redundant information arising from complex image backgrounds and feature interactions, AFEM adaptively adjusts filter parameters, enhancing salient features and suppressing interference, gradually restoring the image's resolution. Finally, MFIM extensively mines multiscale detailed information from low-level features for refining high-level semantics, achieving the inference of high-quality saliency maps. Through extensive and comprehensive experimental evaluations, we have demonstrated the exceptional performance of the proposed ASNet.

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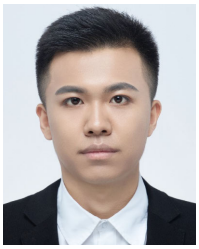
Guohua Geng received the bachelor's and master's degrees from Northwestern University, Evanston, IL, USA, in 1976 and 1988 respectively.

She is a Professor with Northwest University and a Ph.D. Supervisor. Her mainly engaged in theoretical and applied engineering innovation research in the fields of virtual reality, visualization technology, image processing, and intelligent information processing.



Yufei Cao was born in 1999. He is currently pursuing the master's degree in computer technology with Ningxia University, Yinchuan, China.

His main research areas include deep learning and image processing.



Ruixiang Yan received the master's degree in computer technology from Ningxia University, Yinchuan, China, in 2022.

He is currently a Research Assistant at Northwest University, Xi'an, China. His primary research interests include computer vision and image processing.



Pengbo Zhou received the master's degree from the University of Brittany, Rennes, France, in 2010, and the Ph.D. degree from the School of Information and Science and Technology, Beijing Normal University, Beijing, China, in 2020.

He is currently an Engineer at the Experimental Center, School of Art and Media, Beijing Normal University. His research interests include digital art and virtual reality.



Longquan Yan received the master's degree in computer technology from Ningxia University, Yinchuan, China, in 2022. He is currently pursuing the Ph.D. degree in software engineering with Northwest University, Xi'an, China.

His main research interests include remote sensing image segmentation, salient object detection, and image generation.



Yongle Meng received the bachelor's degree from Guangxi University for Nationalities, Nanning, China, in 2020. She is currently pursuing the master's degree with the Xi'an University of Science and Technology, Xi'an, China.

Her research focuses on deep learning and data mining.

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